

Usha Sree Paleboyina

Portfolio | LinkedIn | GitHub

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EDUCATION

- University of Florida** Gainesville, FL
Master of Science, Computer and Information Science Engineering (CGPA: 3.8) Jan 2024 – Dec 2025
- Vignan Lara Institute of Technology and Science** India
Bachelor of Technology, Computer Science Engineering (CGPA: 3.7) Aug 2019 – May 2023

SKILLS

- Programming:** C++, Java, Python, JavaScript, SQL, TypeScript, HTML5, CSS3
- Data Bases:** PostgreSQL, MySQL, DynamoDB
- Backend Tools:** Object-Oriented Programming (OOP), Multithreading, RESTful APIs, Microservices, WebSockets
- Frameworks & Libraries:** Spring Boot, FastAPI, Flask, React.js, NumPy, Pandas, SciPy, Plotly, Matplotlib, Streamlit
- Cloud & DevOps:** AWS (EC2, Lambda, S3, SQS, SNS, Step Functions), Docker, Kubernetes, CI/CD (GitHub Actions), Git
- Machine Learning & AI:** OpenAI API, LangChain, spaCy, scikit-learn, TensorFlow, TF-IDF.
- Testing & UI Tools:** JUnit, PyTest, Postman, Selenium, Unit & Integration Testing

EXPERIENCE

- Student Research Assistant** Gainesville, FL
University of Florida Jul 2024 – Present
 - Designed and implemented RESTful APIs using Flask and FastAPI to serve as the core integration layer for automated simulation workflows.
 - Developed high-throughput backend services leveraging NumPy and SciPy to perform optimized numerical computations on large-scale datasets.
- Programmer Analyst** India
VLITS Mar 2023 – Dec 2023
 - Led backend development of a wholesale management platform using Java and Spring Boot, exposing secure APIs for investment and tax data workflows.
 - Developed an event-driven notification system with AWS Lambda, SQS, and SNS to automate real-time alerts, enhancing communication efficiency by 60%.
- Software Engineering Intern** India
VLITS Aug 2022 – Mar 2023
 - Developed secure RESTful services using Spring Boot and Layer 7 API Gateway for authentication, account creation, and data aggregation.
 - Integrated LDAP authentication and role-based access policies to safeguard sensitive user data and enforce compliance standards.

PROJECTS

- Distributed Music Analytics Platform:**
 - Architected secure data processing platform using Flask, PostgreSQL, and Kubernetes to ingest and process multi-genre music datasets at scale.
 - Implemented modular backend pipelines to transform raw musical sequences into structured representations, automating the extraction of rhythmic patterns.
 - Optimized ML inference services via REST APIs (TensorFlow), achieving sub-second latency for real-time classification across 10k+ audio samples.
- MediScan Web Platform:**
 - Designed and implemented a modular backend system using Java (Spring Boot) and PostgreSQL, serving concurrent user requests through well-defined REST APIs.
 - Architected a service-oriented backend with clear separation of concerns (auth, ingestion, processing, retrieval), enabling independent scaling and maintainable code ownership.
 - Implemented secure authentication and authorization using OAuth 2.0 and JWT, enforcing role-based access control for sensitive operations.
- Sensitive Data Redaction and Crime Prediction Platform:**
 - Engineered RESTful microservices using FastAPI and Java Servlets for real-time PII redaction and entity recognition.
 - Built interactive visualizations with Streamlit and Google Maps API to identify crime hotspots and time-based patterns for decision support.
 - Containerized and deployed services via Docker on AWS EC2, increasing scalability and deployment efficiency by 40%.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional
- C++ by NPTEL